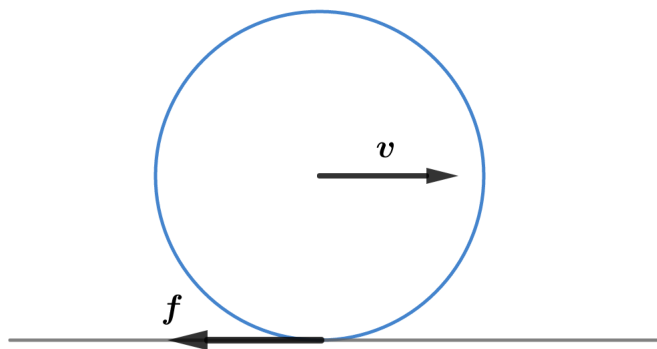


2018B F=ma Exam: Problem 8

Kevin S. Huang



We have

$$f = \mu N = \mu mg$$

so

$$\tau = fR = \mu mgR$$

Since the torque is constant, the ball will have constant angular acceleration until it rolls without slipping. Thus, angular velocity increases linearly and is then constant so the answer is C.